



PROJECT REPORT



NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

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**DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING**

**SUMMER INTERNSHIP ON
IOT AND DATA ANALYTICS FOR SMART SYSTEM DESIGN**

SUMMER INTERNSHIP 2026

PROJECT TITLE:

MOTORIZED CURTAIN SYSTEM

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IoT



Data Analytics



Smart Systems



Automation



Innovation

Abstract :

This project presents a **Motorized Curtain System** using the **ESP32** microcontroller for smart home automation. The system automatically opens and closes the curtain based on different environmental conditions. It uses an **LDR sensor** to detect sunlight, an **FC-37 rain sensor** to detect rain, an **IR sensor** to detect obstacles, and a **DHT11 sensor** to measure temperature and humidity.

When there is enough sunlight, the curtain opens automatically to allow natural light into the room. During rain or at night, the curtain closes to protect the room and provide privacy. The IR sensor stops the curtain movement if it detects any obstacle, ensuring safe operation. Two **28BYJ-48 stepper motors** controlled by **ULN2003 driver modules** move the curtain smoothly, while **limit switches** stop the motors when the curtain reaches the fully open or fully closed position.

The ESP32 processes all sensor data and controls the curtain automatically. Temperature and humidity values are displayed on the Serial Monitor for monitoring. This system helps save energy, improves comfort, reduces manual effort, and increases safety. The project demonstrates a simple, low-cost, and efficient smart curtain solution suitable for homes, offices, hotels, hospitals, and other smart building applications.

Introduction :

In today's world, smart home technology is becoming more popular because it makes daily life easier, safer, and more comfortable. Home automation systems help control different household devices automatically, reducing manual effort and improving convenience. One such application is a **Motorized Curtain System**, which can automatically open and close curtains based on environmental conditions.

The **Motorized Curtain System** is designed using the **ESP32** microcontroller and various sensors such as the **FC-37 Rain Sensor**, **LDR (Light Dependent Resistor)**, **IR Obstacle Sensor**, and **DHT11 Temperature and Humidity Sensor**. The system also uses two **28BYJ-48 stepper motors** with **ULN2003 driver modules** to move the curtains smoothly.

The system works automatically by monitoring the surrounding environment. During the daytime, when sufficient sunlight is available, the curtain opens to allow natural light into the room. At night or when rain is detected, the curtain closes automatically to provide privacy and protect the room from rain. The IR sensor detects obstacles while the curtain is moving and stops the motor to prevent accidents. Limit switches are used to stop the curtain at the fully open and fully closed positions, ensuring safe and reliable operation.

The ESP32 continuously reads data from all the sensors and controls the motors according to predefined conditions. It also displays temperature and humidity values on the Serial Monitor for monitoring purposes.

This project provides a simple, low-cost, and energy-efficient solution for smart home automation. It reduces manual work, improves user comfort, increases safety, and helps save energy by making better use of natural sunlight. The system can be used in homes, offices, hotels, hospitals, schools, and other buildings where automatic curtain control is beneficial.

Objectives :

The main objective of this project is to design and develop a **Motorized Curtain System** using the **ESP32** microcontroller that can automatically open and close curtains based on environmental conditions. The system aims to improve comfort, safety, and energy efficiency while reducing manual effort.

Objectives

- To design and implement an automatic curtain control system using the ESP32 microcontroller.
- To automatically open the curtain when sufficient sunlight is detected using the LDR sensor.
- To automatically close the curtain when rain is detected using the FC-37 rain sensor.
- To detect obstacles using the IR sensor and stop the curtain movement for safe operation.
- To monitor and display temperature and humidity using the DHT11 sensor.

- To control two stepper motors simultaneously for smooth curtain movement.
- To use limit switches to stop the curtain at fully open and fully closed positions.
- To reduce manual effort by automating curtain operation.
- To improve energy efficiency by making better use of natural sunlight.
- To provide a low-cost, reliable, and user-friendly smart home automation solution.
- To develop a system that can be used in homes, offices, hotels, hospitals, schools, and other smart buildings.

Hardware Requirements :

The **Motorized Curtain System** is developed using the **ESP32 microcontroller** and different sensors to automate the opening and closing of curtains. The hardware components used in this project are listed below.

Sl. No.	Component	Quantity	Purpose
1	ESP32 DevKit V1	1	Main controller that processes sensor data and controls the system.
2	28BYJ-48 Stepper Motor	2	Used to open and close the curtain smoothly.

Sl. No.	Component	Quantity	Purpose
3	ULN2003 Stepper Motor Driver	2	Drives the stepper motors based on ESP32 commands.
4	FC-37 Rain Sensor	1	Detects rain and closes the curtain automatically.
5	LDR (Light Dependent Resistor) Module	1	Detects sunlight and controls curtain opening and closing.
6	IR Obstacle Sensor	1	Detects obstacles and stops the curtain movement for safety.
7	DHT11 Temperature & Humidity Sensor	1	Measures room temperature and humidity.
8	Limit Switch	2	Stops the curtain at fully open and fully closed positions.
9	Breadboard	1	Used for making temporary circuit connections.
10	Jumper Wires	As Required	Connects all components together.
11	5V Power Supply	1	Provides power to the stepper motors and driver

Sl. No.	Component	Quantity	Purpose
			modules.
12	USB Cable	1	Used to program and power the ESP32.

Hardware Description

1. ESP32 DevKit V1

The ESP32 is the main controller of the project. It reads data from all the sensors and controls the stepper motors to open or close the curtain automatically.

2. 28BYJ-48 Stepper Motor

Two stepper motors are used to move the curtain smoothly. They rotate in both clockwise and anticlockwise directions according to the control signals from the ESP32.

3. ULN2003 Motor Driver

The ULN2003 driver acts as an interface between the ESP32 and the stepper motors. It provides the required current to drive the motors safely.

4. FC-37 Rain Sensor

The FC-37 rain sensor detects rainwater. When rain is detected, it sends a signal to the ESP32, which automatically closes the curtain.

5. LDR Module

The LDR (Light Dependent Resistor) measures the amount of light. When sufficient sunlight is available, the curtain opens automatically. At night or in low light, the curtain closes.

6. IR Obstacle Sensor

The IR sensor detects objects in front of the moving curtain. If an obstacle is detected, the motor stops immediately to prevent accidents.

7. DHT11 Sensor

The DHT11 sensor measures the room temperature and humidity. The readings are displayed on the Serial Monitor for monitoring.

8. Limit Switches

Two limit switches are used to detect the fully open and fully closed positions of the curtain. They prevent the motors from moving beyond the safe limits.

9. Breadboard

The breadboard is used to connect all electronic components without soldering, making the circuit easy to build and modify.

10. Jumper Wires

Jumper wires are used to connect the ESP32, sensors, motor drivers, and other components.

11. 5V Power Supply

A separate 5V power supply provides enough power to the stepper motors and ULN2003 driver modules for stable operation.

Software Requirements :

The **Motorized Curtain System** is developed using the **Arduino IDE** and programmed in **C++**. The software is responsible for reading sensor data, processing the information, and controlling the stepper motors automatically. The software also displays temperature, humidity, and curtain status on the Serial Monitor.

Software Requirements Table

Sl. No.	Software	Purpose
1	Arduino IDE	Used to write, edit, compile, and upload the program to the ESP32.
2	ESP32 Board Package	Enables Arduino IDE to program the ESP32 microcontroller.
3	C++ Programming Language	Used to write the program for the ESP32.

Sl. No.	Software	Purpose
4	Stepper Library	Controls the 28BYJ-48 stepper motors.
5	DHT Sensor Library	Reads temperature and humidity values from the DHT11 sensor.
6	Adafruit Unified Sensor Library	Supports the DHT sensor library and improves sensor compatibility.
7	Serial Monitor	Displays sensor readings and system status during operation.
8	Windows 11	Operating system used for programming and testing the project.

Software Description

1. Arduino IDE

The Arduino IDE is the main software used to write, edit, compile, and upload the program to the ESP32. It also provides the Serial Monitor to display sensor values and system status.

2. ESP32 Board Package

The ESP32 Board Package is installed in the Arduino IDE to support programming and uploading code to the ESP32 development board.

3. C++ Programming Language

The project is programmed using C++. It allows the ESP32 to process sensor inputs and control the motors automatically.

4. Stepper Library

The Stepper library is used to control the two 28BYJ-48 stepper motors. It manages motor speed, direction, and smooth curtain movement.

5. DHT Sensor Library

This library is used to read temperature and humidity data from the DHT11 sensor accurately.

6. Adafruit Unified Sensor Library

This library works together with the DHT Sensor Library and provides a common interface for reading sensor data.

7. Serial Monitor

The Serial Monitor displays real-time information such as:

- Rain Status
- Light Level
- Obstacle Detection
- Temperature
- Humidity
- Curtain Status (Open, Closed, Opening, Closing, or Stopped)

8. Windows 11

Windows 11 is used as the operating system for developing, uploading, and testing the project.

Methodology :

The **Motorized Curtain System** is designed to automatically open and close curtains based on environmental conditions. The system uses the **ESP32 microcontroller** to collect data from different sensors and control the stepper motors. The complete working process is explained below.

Step 1: System Initialization

When the power supply is turned on, the ESP32 initializes all the connected components, including the FC-37 Rain Sensor, LDR, IR Sensor, DHT11 Sensor, Stepper Motors, and Limit Switches. The system then starts monitoring the sensor values continuously.

Step 2: Reading Sensor Data

The ESP32 continuously reads data from all the sensors.

- **FC-37 Rain Sensor** checks whether it is raining.
- **LDR Sensor** measures the amount of sunlight.
- **IR Sensor** detects any obstacle in the curtain path.
- **DHT11 Sensor** measures the room temperature and humidity.

- **Limit Switches** detect whether the curtain is fully open or fully closed.
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Step 3: Processing the Sensor Data

The ESP32 compares the sensor readings with predefined conditions.

- If rain is detected, the curtain should close.
 - If there is enough sunlight and no rain, the curtain should open.
 - If it is dark, the curtain should close.
 - If an obstacle is detected, the curtain movement should stop immediately.
-

Step 4: Controlling the Stepper Motors

Based on the sensor data, the ESP32 sends control signals to the two ULN2003 motor driver modules.

- Both stepper motors rotate together to move the curtain.
 - The curtain opens or closes smoothly depending on the conditions.
 - The motors stop automatically when a limit switch is activated.
-

Step 5: Displaying Sensor Values

The ESP32 displays the following information on the Arduino Serial Monitor:

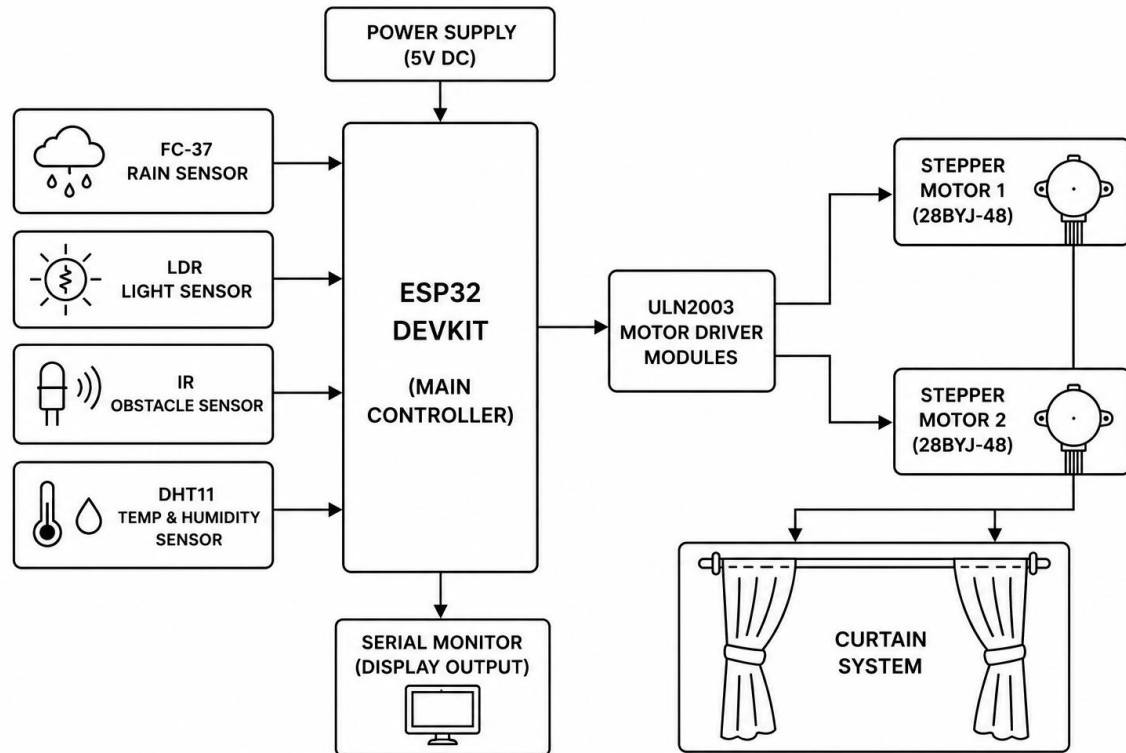
- Rain Status
- Light Level
- Obstacle Status
- Temperature
- Humidity
- Curtain Status (Opening, Closing, Open, Closed, or Stopped)

This helps monitor the system in real time.

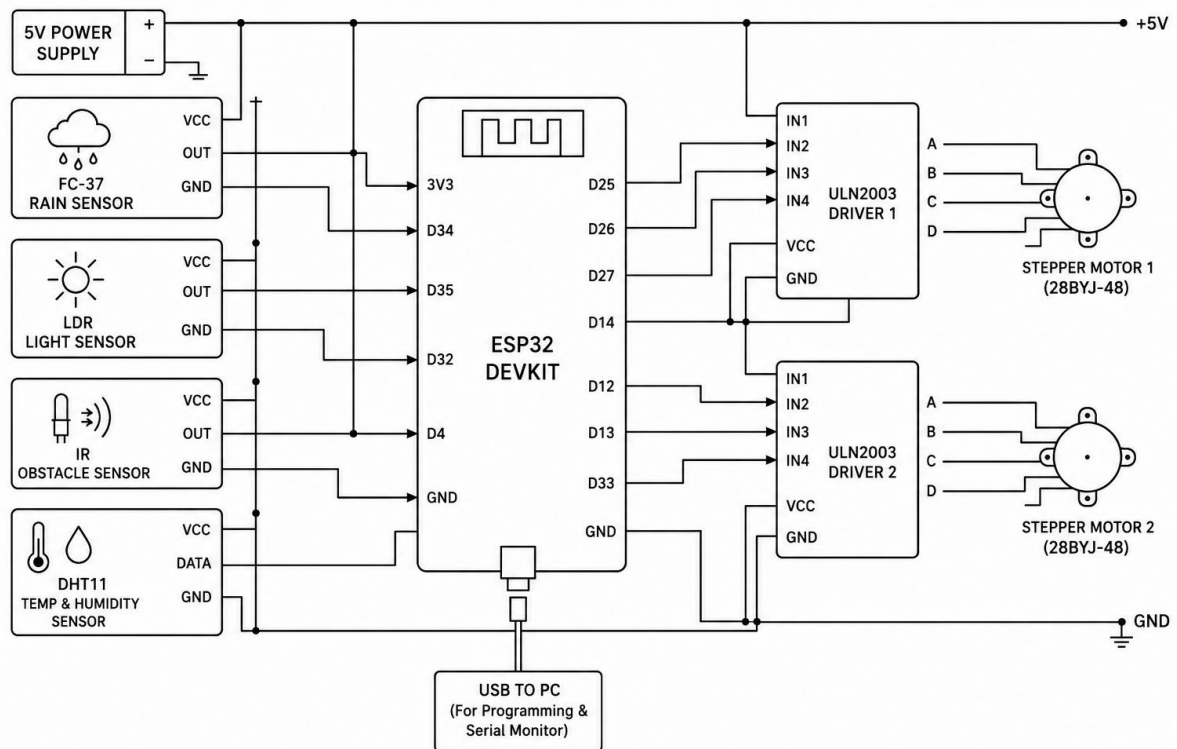
Step 6: Continuous Monitoring

After completing one cycle, the ESP32 continues reading the sensor values and updates the curtain position whenever environmental conditions change. This process repeats continuously to ensure automatic and reliable operation.

Block Diagram :



Circuit Diagram :



Working Principle :

The Motorized Curtain System is an automatic curtain control system that uses the ESP32 microcontroller and different sensors to open and close the curtain based on environmental conditions. The ESP32 continuously monitors the sensor readings and controls the stepper motors automatically.

Step 1: System Initialization

When the power supply is turned on, the ESP32 starts the program and initializes all the connected sensors and stepper motors. After initialization, the system begins monitoring the surroundings.

Step 2: Rain Detection

The FC-37 Rain Sensor continuously checks for rain.

- If rain is detected, the ESP32 immediately sends a signal to the stepper motors to close the curtain.**
 - If there is no rain, the system checks the light level using the LDR sensor.**
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Step 3: Light Detection

The LDR (Light Dependent Resistor) Sensor measures the intensity of sunlight.

- If sufficient sunlight is available, the ESP32 opens the curtain to allow natural light into the room.
 - If it becomes dark, the ESP32 closes the curtain automatically.
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Step 4: Obstacle Detection

The IR Obstacle Sensor continuously monitors the curtain path.

- If a person or object is detected while the curtain is moving, the ESP32 immediately stops both stepper motors.
- After the obstacle is removed, the curtain resumes its movement.

This feature improves the safety of the system.

Step 5: Temperature and Humidity Monitoring

The DHT11 Sensor measures the room temperature and humidity.

- The ESP32 reads the sensor values every second.
 - The temperature and humidity readings are displayed on the Arduino Serial Monitor for monitoring.
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Step 6: Curtain Movement

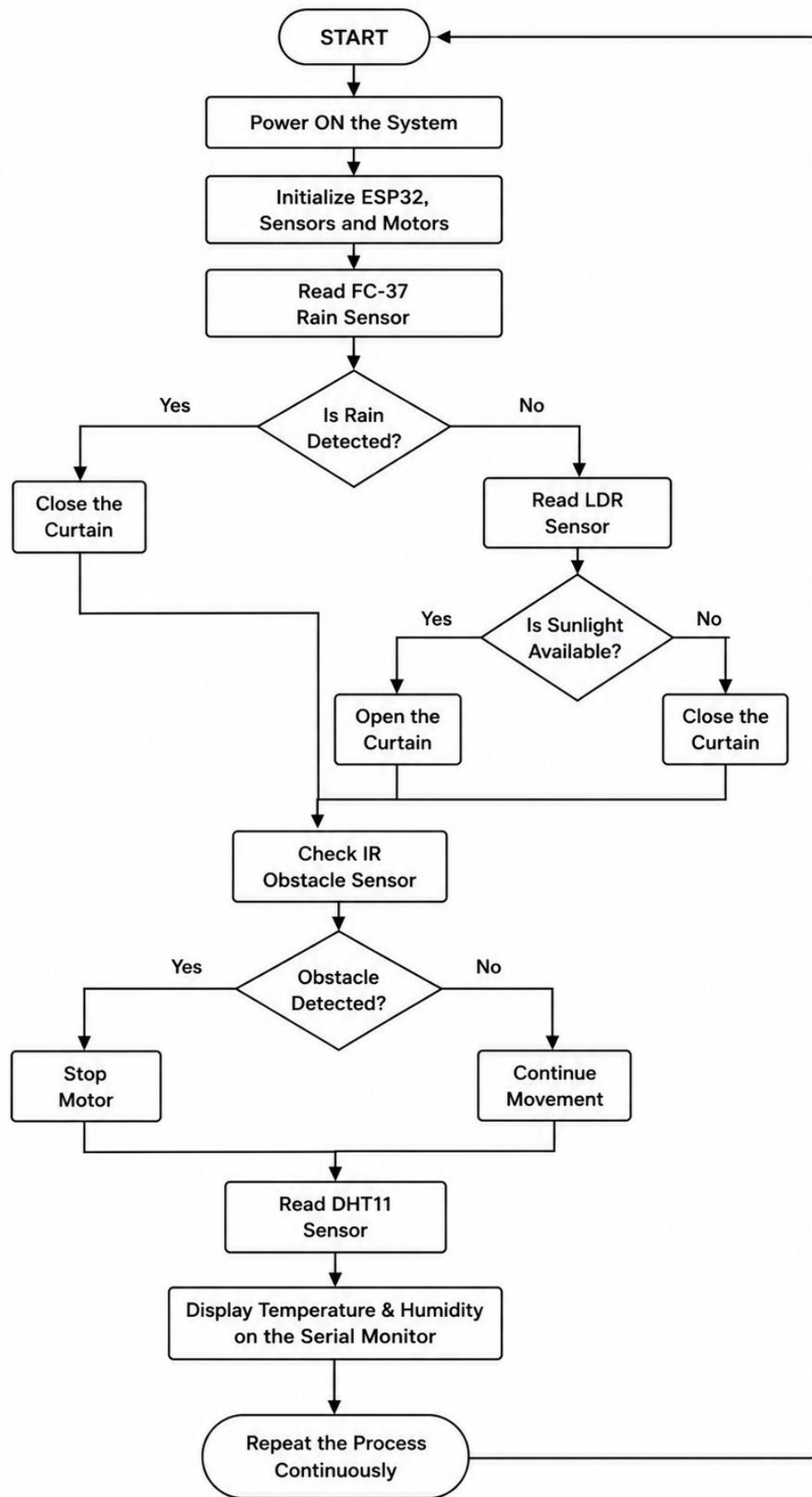
The ESP32 controls two 28BYJ-48 stepper motors through the ULN2003 motor driver modules.

- **Both motors rotate together to move the curtain smoothly.**
 - **The curtain opens or closes depending on the sensor readings.**
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Step 7: Continuous Operation

The ESP32 continuously reads all sensor values and updates the curtain position whenever the environmental conditions change. This process repeats continuously, allowing the system to operate automatically without user intervention.

Working Flow :



Results

The **Motorized Curtain System** was successfully designed and tested using the **ESP32 microcontroller**, **FC-37 Rain Sensor**, **LDR Sensor**, **IR Obstacle Sensor**, **DHT11 Sensor**, **ULN2003 Motor Drivers**, and **two 28BYJ-48 Stepper Motors**. The system operated according to the programmed conditions and performed all the required functions correctly.

Observed Results

- The curtain **opened automatically** when sufficient sunlight was detected by the LDR sensor.
- The curtain **closed automatically** when the FC-37 rain sensor detected rain.
- The curtain also **closed automatically** when the light level became low (night condition).
- The **IR obstacle sensor** successfully detected obstacles and immediately stopped the motors to ensure safe operation.
- After the obstacle was removed, the curtain resumed normal operation.
- The **DHT11 sensor** measured room temperature and humidity correctly and displayed the values on the Arduino Serial Monitor every second.
- The two **28BYJ-48 stepper motors** rotated together smoothly, ensuring proper curtain movement.

- The ESP32 processed all sensor data accurately and controlled the motors without any noticeable delay.
- The system operated continuously and reliably under different environmental conditions.

Serial Monitor Output

The Arduino Serial Monitor displayed the following information in real time:

- Rain Status
- Light Level
- IR Sensor Status
- Temperature (°C)
- Humidity (%)
- Curtain Status (Opening, Closing, Open, Closed, or Stopped)

Advantages :

The **Motorized Curtain System** provides a smart and automatic way to control curtains. It improves comfort, safety, and energy efficiency while reducing manual effort. The main advantages of this project are listed below.

Advantages

- Automatically opens and closes the curtain based on sunlight and rain.

- Reduces manual effort by eliminating the need to operate the curtain manually.
- Improves comfort by allowing natural sunlight into the room during the day.
- Provides privacy by closing the curtain automatically at night or during rain.
- Saves electrical energy by making better use of natural daylight.
- Enhances safety by stopping the curtain when an obstacle is detected using the IR sensor.
- Displays room temperature and humidity in real time using the DHT11 sensor.
- Provides smooth and reliable curtain movement using two stepper motors.
- Uses the ESP32 microcontroller, which offers fast processing and reliable performance.
- Low-cost and easy to build using commonly available electronic components.
- Simple to operate and requires very little user intervention.
- Suitable for smart home and IoT-based automation applications.
- Can be expanded with Wi-Fi, Bluetooth, or a mobile application for remote control in the future.

- Consumes low power and is suitable for continuous operation.
- Easy to maintain and upgrade with additional sensors or smart features.

Future Scope :

The **Motorized Curtain System** can be improved by adding more smart features and advanced technologies. These improvements will make the system more intelligent, convenient, and suitable for modern smart homes.

Future Scope

- Control the curtain remotely using a **mobile application**.
- Add **Wi-Fi connectivity** to monitor and control the curtain from anywhere using the Internet.
- Integrate with **Google Assistant** or **Amazon Alexa** for voice control.
- Develop a **web dashboard** to monitor sensor values and curtain status in real time.
- Store sensor data on a **cloud platform** for analysis and monitoring.
- Send notifications to the user's mobile phone when the curtain opens, closes, or when rain is detected.

- Add a **Real-Time Clock (RTC)** module to open and close the curtain at scheduled times.
- Use **solar power** to make the system more energy-efficient and eco-friendly.
- Replace the DHT11 with more accurate sensors such as the **DHT22** or **BME280** for improved environmental monitoring.
- Add an **LCD or OLED display** to show temperature, humidity, and curtain status directly on the device.
- Improve obstacle detection using **ultrasonic** or **camera-based sensors** for better safety.
- Integrate **Artificial Intelligence (AI)** to automatically learn user preferences and control the curtain more efficiently.
- Connect the system with other **smart home devices** such as smart lights, fans, and air conditioners.
- Improve the motor mechanism to support **heavier curtains** and larger windows.
- Enhance security by adding **password protection** or **biometric authentication** for manual control.

• **Conclusion :**

- The **Motorized Curtain System** is a simple and efficient smart home automation project developed using the **ESP32 microcontroller**. The system automatically opens and closes the curtain based on environmental conditions such as **sunlight** and **rain**. It uses an **LDR sensor** to detect light, an **FC-37 rain sensor** to detect rain, an **IR sensor** to

detect obstacles, and a **DHT11 sensor** to monitor temperature and humidity.

- The project successfully performs automatic curtain control without requiring manual operation. The **IR sensor** improves safety by stopping the curtain whenever an obstacle is detected, while the **DHT11 sensor** continuously displays temperature and humidity on the Serial Monitor. The two **28BYJ-48 stepper motors** work together to provide smooth and reliable curtain movement.
- This project helps improve comfort, privacy, and energy efficiency by making better use of natural sunlight and protecting the room during rain. It is low-cost, easy to use, and suitable for homes, offices, hotels, hospitals, schools, and other smart building applications.
- Overall, the **Motorized Curtain System** demonstrates how IoT and automation technologies can make everyday life easier and more convenient. It provides a reliable foundation for future improvements such as mobile app control, Wi-Fi connectivity, cloud monitoring, and voice assistant integration, making it a useful solution for modern smart homes.

References :

The following books, websites, and datasheets were referred to during the design and development of the **Motorized Curtain System** project.

Books

1. Simon Monk, **Programming Arduino: Getting Started with Sketches**, McGraw-Hill Education.
2. Michael Margolis, **Arduino Cookbook**, O'Reilly Media.

3. Charles Platt, **Make: Electronics**, Maker Media.

Official Websites

1. **Arduino Official Website** – <https://www.arduino.cc>
2. **ESP32 Documentation (Espressif Systems)** – <https://docs.espressif.com>
3. **Arduino Library Manager** – <https://www.arduino.cc/reference/en/libraries/>

Datasheets

1. ESP32 DevKit V1 Datasheet
2. DHT11 Temperature and Humidity Sensor Datasheet
3. FC-37 Rain Sensor Datasheet
4. LDR (Light Dependent Resistor) Datasheet
5. IR Obstacle Sensor Datasheet
6. 28BYJ-48 Stepper Motor Datasheet
7. ULN2003 Stepper Motor Driver Datasheet

Online Resources

1. Arduino Tutorials and Examples
2. ESP32 Programming Guides
3. IoT and Smart Home Automation Tutorials
4. Electronics Circuit Design References
